I. From bits to qubits

- classical states for computation are either "0" or "1".
- in quantum mechanics a state can be in superposition, i.e, simultaneously in "0" and "1"
  - ↳ superpositions allow to perform calculations on many states at the same time
    - => quantum algorithms with exponential speed-up

BUT: once we measure the superposition state, it collapse to one of its states.

=> it is not THAT easy to design quantum algorithms.

Dirac notation

- used to describe quantum states. Let $a, b \in \mathbb{C}^2$
  - ket $|a\rangle = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$.
  - bra $\langle b| = |b\rangle^\dagger = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}^\dagger = (b_1^* \; b_2^*)$

$b = c + d \cdot i$
$b^* = c - di$ $\quad c, d \in \mathbb{R}$

  - bra-ket $\langle b|a\rangle = a_1 b_1^* + a_2 b_2^* = \langle a|b\rangle^* \in \mathbb{C}$

    inner product

  - ket-bra $|a\rangle\langle b| = \begin{pmatrix} a_1 b_1^* & a_1 b_2^* \\ a_2 b_1^* & a_2 b_2^* \end{pmatrix}$

    $|a\rangle\langle b|$

- we define the states $|0\rangle := \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ which are orthogonal

$$\langle 0|1\rangle = (1 \; 0) \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0.$$

- all quantum states are normalized, i.e $\langle \psi|\psi\rangle = 1$.

e.g $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$

Normalized.
Physically it has no meaning, just a convention.

Measurements

- we choose orthogonal bases to describe measure quantum states.
- during a measurement onto the basis $\{|0\rangle, |1\rangle\}$, the state will collapse into either of the states. (either $|0\rangle$ or $|1\rangle$)
  - => as these are eigenstates of (sigma-z i) σ_z, we call this a z-measurement.

- theoretically there are infinitely many different bases, but other common ones are

$$|+\rangle := \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle := \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

orthogonal

$$|+i\rangle := \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle) \qquad |-i\rangle := \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$$

corresponding to the eigenstates of σ_x and σ_y, respectively.

- Born rule: the probability that a state $|\psi\rangle$ collapse during a projective measurement onto the basis $\{|x\rangle, |x^\perp\rangle\}$ to the state $|x\rangle$ is given by

$$\mathbb{P}(x) = |\langle x|\psi\rangle|^2, \qquad \sum_i \mathbb{P}(x_i) = 1.$$

Examples: ① $|\psi\rangle = \frac{1}{\sqrt{3}}(|0\rangle + \sqrt{2}|1\rangle)$ is measured in the basis $\{|0\rangle, |1\rangle\}$.

$$\rightarrow \mathbb{P}(0) = \left|\langle 0|\frac{1}{\sqrt{3}}(|0\rangle + \sqrt{2}|1\rangle)\right|^2 =$$

$$= \left|\frac{1}{\sqrt{3}}\underbrace{\langle 0|0\rangle}_{1} + \frac{\sqrt{2}}{3}\underbrace{\langle 0|1\rangle}_{0}\right|^2 = \frac{1}{3}.$$

$$\Rightarrow \mathbb{P}(1) = 2/3.$$

② $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ is meas. in the basis $\{|+\rangle, |-\rangle\}$:

$$\hookrightarrow \mathbb{P}(+) = \left|\langle +|\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right|^2 =$$

$$= \frac{1}{2}\cdot\frac{1}{2}\left|(|0\rangle + |1\rangle)(|0\rangle - |1\rangle)\right|^2 \circledast$$

$$\circledast \frac{1}{4}\left|\langle 0|0\rangle - \langle 0|1\rangle + \langle 1|0\rangle - \langle 1|1\rangle\right|^2 =$$

$$= \frac{1}{4}|1 - 0 + 0 - 1|^2 = 0. \Rightarrow \mathbb{P}(|-\rangle) = 1.$$

Bloch Sphere

We can write any normalized (pure) state as:

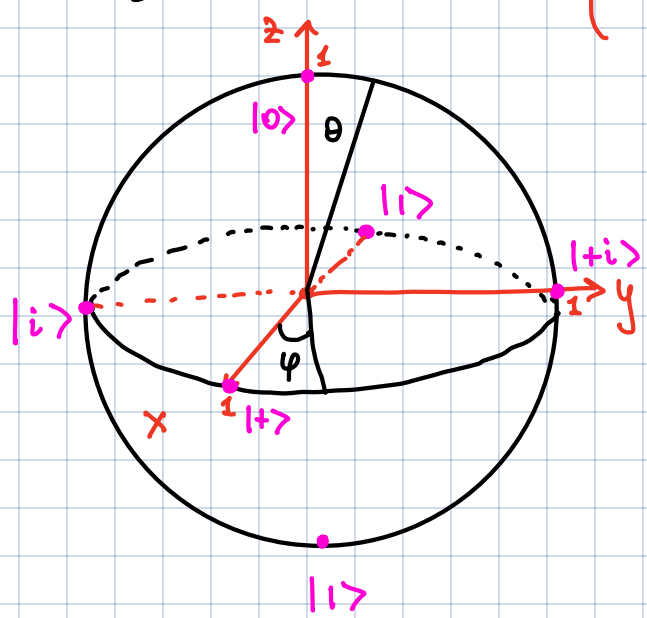
$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle$$

where $\varphi \in [0, 2\pi)$ describes the relative phase.

and $\theta \in [0, \pi]$ determines the probability to measure $|0\rangle / |1\rangle$ $\quad p(0) = \cos^2\frac{\theta}{2}, p(1) = \sin^2\frac{\theta}{2}$

→ all normalized pure states can be illustrated on the surface of a sphere with radius $|\vec{r}| = 1$, which we call the Bloch Sphere.

⇒ the coordinates of such a state are given by the Bloch vector: $\vec{r} \begin{pmatrix} \sin\theta\cos\theta \\ \sin\theta\sin\varphi \\ \cos\theta \end{pmatrix}$



Examples:

$|0\rangle$: $\theta = 0$, φ arbitrary

$\rightarrow \vec{r} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$|1\rangle$: $\theta = \pi$, φ arbitrary

$\rightarrow \vec{r} = \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix}$

$|+\rangle$: $\theta = \frac{\pi}{2}$, $\varphi = 0 \rightarrow \vec{r} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$|-\rangle$: $\theta = \frac{\pi}{2}$, $\varphi = \pi \rightarrow \vec{r} = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$

$|+i\rangle$: $\theta = \frac{\pi}{2}$, $\varphi = \frac{\pi}{2} \rightarrow \vec{r} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

$|-i\rangle$: $\theta = \frac{\pi}{2}$, $\varphi = \frac{3\pi}{2} \rightarrow \vec{r} = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}$

Be careful: On the Bloch sphere angles are twice as big as in Hilbert space, e.g. $|0\rangle$ & $|1\rangle$ are orthogonal, but on the Bloch sphere their angle is $180°$.

For a general state $|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + \ldots$

→ θ is the angle on the Bloch sphere.

While $\frac{\theta}{2}$ is the actual angle in Hilbert space!

$\Rightarrow$ z-measurement corresponds to a projection onto the z-axis and analagously for x & y.